Clothing Size Recommendation Using Body Measurement Estimation

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Abstract. Incorrect apparel sizing in online shopping leads to high return rates, customer dissatisfaction, and environmental waste. Traditional methods, such as manual measurements, are impractical for remote shoppers. This project presents the clothing size recommendation system, which requires users to input their height and capture two images (front and side) with guided on-screen overlays to ensure proper alignment. The system employs YOLOv11 model to extract key body points from both images, which, when combined with the user's height through scaling techniques, yield accurate estimations of 3D body dimensions such as shoulder width and belly circumference. These measurements are then fed into multiple machine learning models for size classification. After comprehensive evaluation of four different algorithms—Logistic Regression, Random Forest, Support Vector Machine, and XGBoost—we identified Random Forest as the optimal model, achieving 97.50% accuracy in clothing size prediction.

Keywords: Clothing size recommendation · Body measurement estimation · Computer vision · Pose estimation · YOLOv11 · Random Forest

1 Introduction

Clothing size is a common issue, especially for online shoppers or those unaware of their body dimensions, often leading to incorrect purchases and increased return rates [1, ?]. Traditional manual measurement methods struggle to capture individual body shape variations and are inconvenient for online use. To address this, we propose a clothing size recommendation system using a multi-image method to estimate body measurements for personalized size suggestions. Users provide their height and capture two images—front and side—with guided overlays to ensure proper alignment. This dual-view approach improves estimation of key dimensions like shoulder width and waist circumference [2]. Using YOLOv11 for pose estimation, 2D landmarks are extracted and input into multiple machine learning models—Logistic Regression, Random Forest, SVM, and XGBoost—to determine the best size classification approach. This method bridges the gap between simple automated methods and traditional manual accuracy. In this study, we present the image capture and processing techniques, system design and implementation, a comparative analysis of models, and an evaluation of real-world performance. This project's main contributions are:

- The creation of an intuitive user interface that makes it easier to take precise front and side view photos.
- The use of pose estimation techniques to extract important body landmarks.
- The reconstruction of 3D body measurements using height-based scaling.
- A comprehensive comparison of four machine learning models for clothing size classification.
- The identification of Random Forest as the optimal model for size prediction with 97.50% accuracy.
- The integration of these measurements with an extensive database of clothing sizes to provide customized size recommendations.

The remainder of this paper is organized as follows: Section 2 presents the project architecture and frameworks, including image processing, human pose detection, and body measurement calculation. Section 3 reports the results, evaluating measurement accuracy and limitations. Section 4 discusses planned improvements and future work to enhance measurement and size recommendation accuracy. The final section concludes the paper, summarizing our work and its potential impact on the online fashion retail industry.

2 Related Work

Recent studies have applied YOLO for human detection, logistic regression for classification, and various methods for clothing size estimation. YOLO-based approaches excel in pose detection, while logistic regression provides interpretability, and fashion applications benefit from both 2D and 3D measurement techniques. However, existing works often lack integration, either relying on expensive equipment or suffering from limited accuracy. Our work combines pose estimation and machine learning into a practical pipeline that balances accuracy and accessibility.

2.1 YOLO-Based Human Detection and Pose Estimation

The YOLO family has become a key framework for real-time object detection due to its balance of speed and accuracy. Recent variants, such as CST-YOLO [3] and YOLOv7-tiny [4], improve performance on small objects and resource-constrained devices, while others like improved YOLOv7 [5] and YOLO-RSFM [6] enhance robustness against occlusion and scale variation. These works demonstrate YOLO's adaptability beyond traditional detection tasks. However, most focus on generic detection rather than anthropometric estimation. Our work leverages YOLOv11 for pose landmark extraction, enabling accurate body measurement from multi-view images.

2.2 Logistic Regression and Statistical Modeling

Parallel to deep learning advances, logistic regression remains a reliable method for classification due to its interpretability and efficiency. Recent studies have improved its robustness with enriched features [7] and explored model selection strategies [8]. In fashion applications, regression has been used to predict missing body measurements [9], showing its continued relevance. However, most works are theoretical or general-purpose, with limited integration into vision-based pipelines. In our study, logistic regression serves as the final mapping stage, translating extracted anthropometric features into discrete clothing size categories.

2.3 Clothing Size Estimation and Fashion Applications

The fashion industry has increasingly explored body size prediction and clothing recommendation systems. One approach uses 2D images with CNN-based methods [10], combining body contour and keypoint detection, which achieve moderate accuracy but are sensitive to pose, distance, and clothing. Another relies on 3D body scanning [11], offering precise measurements and body shape clustering, but requiring specialized equipment and limiting scalability. Hybrid methods combining pose estimation with limited anthropometric data have been proposed, yet often fail to meet e-commerce accuracy standards. Balancing practicality and precision remains a key challenge.

2.4 Summary

Overall, prior research demonstrates the individual strengths of YOLO-based detection models, logistic regression, and specialized size estimation systems. However, these approaches often operate in isolation: YOLO is rarely applied to measurement estimation, logistic regression is usually studied in abstract settings, and size estimation methods either rely on costly hardware or suffer from limited accuracy in 2D environments. Our work contributes by integrating these streams into a unified pipeline. Specifically, we adopt YOLOv11 for robust multi-view pose landmark extraction and apply logistic regression to map estimated body dimensions to standardized clothing sizes. This design achieves a practical balance between accuracy and accessibility, offering a scalable solution for real-world online shopping platforms.

3 Background

The rise of online shopping creates demand for accurate body size measurement systems. Core methods include computer vision techniques for pose detection, scaling approaches for body dimension estimation, and machine learning for classification. YOLOv11 is used for fast and precise keypoint detection, while algorithms like Logistic Regression, Random Forest, SVM, and XGBoost provide different classification strategies. Random Forest, in particular, is robust to noise and non-linearities, making it suitable for mapping body features to clothing sizes.

3.1 Core Definitions and Concepts

Human Pose Detection This is a part of computer vision that involves recognizing and roughly locating the major joints of the body (e.g., shoulders, elbows, waist, etc.) from images or video streams. Pose generation techniques, such as those implemented in YOLOv11, use convolutional neural networks (CNNs) to predict a set of heatmaps or keypoint maps, each corresponding to one anatomical landmark. By training on large datasets of annotated human poses, the network learns to extract spatial features—like limb orientation, joint appearance, and body silhouette—and to output precise coordinate estimates for each joint in real time. These techniques are useful because:

- **Real-time feedback:** The high processing speed of single-stage CNNs (one forward pass per frame) enables applications such as live fitness coaching, gesture-based interfaces, and interactive gaming without perceptible lag.
- **Robustness to variation:** Deep convolutional filters automatically learn to handle different viewpoints, clothing, lighting, and partial occlusions, making pose detection reliable across diverse environments.
- **Downstream analytics:** Exact joint coordinates allow computation of angles, distances, and movement trajectories, which power higher-level tasks like activity recognition, ergonomic assessment, rehabilitation monitoring, and sports performance analysis.
- **Personalization:** In our clothing-size recommendation system, precise measurements (e.g., shoulder width, arm length) extracted from live video ensure that suggested sizes match each user's unique body proportions, reducing returns and improving customer satisfaction.

YOLOv11 YOLO ("You Only Look Once") is a real-time object detection system using convolutional neural networks, known for its speed and efficiency on both high-end and mobile devices [12]. Although originally for object detection, YOLO can be adapted for tasks like human pose estimation. In our work, a YOLO variant is used to extract precise body measurements from a live camera stream to generate personalized clothing size recommendations.

Logistic Regression Logistic regression is a core statistical classification method that models the probability of a categorical outcome given one or more predictor variables. In our system, we use it to map continuous body measurements (e.g., shoulder width, waist circumference) into discrete clothing size categories (Small, Medium, Large, etc.).

Binary Logistic Function. For the simplest case of two classes (e.g. "Medium" vs. "Not Medium"), the model assumes

$$P(Y = 1 \mid \mathbf{x}) = \sigma(z) = \frac{1}{1 + e^{-z}}, \tag{1}$$

where

$$z = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n, \tag{2}$$

where

- $\mathbf{x} = (x_1, \ldots, x_n)$ is the feature vector of measured dimensions.
- β_0 is the intercept (bias) term, shifting the decision boundary.
- β_i is the weight corresponding to feature x_i, indicating how strongly x_i influences the log-odds.
- $\sigma(\cdot)$ is the logistic (sigmoid) function, which "squashes" real valued z into the interval $(0, 1)$.

We interpret $P(Y = 1 \mid \mathbf{x}) > 0.5$ (or another chosen threshold) as predicting class "1."

Multinomial Extension. To handle $K > 2$ size categories simultaneously, we generalize to

$$P(Y = k \mid \mathbf{x}) = \frac{\exp\left(\beta_{k0} + \boldsymbol{\beta}_k^T \mathbf{x}\right)}{\sum_{j=1}^{K} \exp\left(\beta_{j0} + \boldsymbol{\beta}_j^T \mathbf{x}\right)}, \quad k = 1, \ldots, K, \tag{3}$$

where each class k has its own parameter vector $(\beta_{k0}, \boldsymbol{\beta}_k)$. The predicted size is the k with the highest posterior probability.

Parameter Estimation. All parameters $\{\beta_{k0}, \boldsymbol{\beta}_k\}$ are learned by maximizing the regularized log-likelihood on a labeled training set:

$$\mathcal{L}(\boldsymbol{\beta}) = \sum_{i=1}^{N} \sum_{k=1}^{K} \mathbb{I}(y^{(i)} = k) \, \ln P(Y = k \mid \mathbf{x}^{(i)}) - \frac{\lambda}{2} \sum_{k=1}^{K} \|\boldsymbol{\beta}_k\|_2^2, \tag{4}$$

where λ controls the strength of L_2 regularization to prevent overfitting.

Interpretation of Parameters.

- A positive coefficient β_{ki} means that increasing feature x_i raises the log odds of class k relative to the baseline.
- The magnitude $|\beta_{ki}|$ reflects how strongly x_i influences the probability for size k.
- The intercept β_{k0} shifts the model's bias toward or away from class k when all $x_i = 0$ (after normalization).

In our application, the input $\mathbf{x}$ is the scaled set of body measurements produced by YOLOv11, and the trained logistic model outputs—for each size category—the probability that the measurements belong to that category. We then recommend the size with maximum probability, optionally enforcing a minimum confidence threshold (e.g. 0.7) to prompt the user for retakes if the classification is uncertain.

Random Forest Random Forest [13] is an ensemble learning algorithm that builds a collection of decision trees and aggregates their predictions to improve generalization. Each tree is trained on a bootstrap sample of the training data, and at each split only a random subset of features is considered. This randomness reduces correlation between trees, leading to lower variance compared to a single decision tree. For classification, the final prediction is obtained by majority voting across all trees:

$$\hat{y} = \mathrm{mode}\{h_1(\mathbf{x}), h_2(\mathbf{x}), \ldots, h_T(\mathbf{x})\}, \tag{5}$$

where $h_t(\mathbf{x})$ is the class prediction of tree t, and T is the total number of trees. Random Forests are robust to noise, can capture nonlinear feature interactions, and naturally handle multi-class classification. In our context, they provide a strong nonparametric baseline for mapping anthropometric features into clothing size categories.

Support Vector Machine (SVM) Support Vector Machine [14] is a margin-based classifier that seeks a hyperplane separating classes with maximum margin. For linearly separable data, the decision function is

$$f(\mathbf{x}) = \mathrm{sign}\big(\mathbf{w}^T\mathbf{x} + b\big), \tag{6}$$

where $\mathbf{w}$ is the weight vector orthogonal to the hyperplane and b is the bias. The optimization maximizes the margin $\frac{2}{\|\mathbf{w}\|}$ subject to classification constraints. To handle nonlinearity, kernel functions $\kappa(\mathbf{x}_i, \mathbf{x}_j)$ map inputs into a higher-dimensional space, allowing separation by a linear hyperplane there. Common kernels include polynomial and radial basis function (RBF). For multi-class size classification, strategies such as one-vs-one (OvO) or one-vs-rest (OvR) are applied. SVMs are effective with high-dimensional data and small training sets, making them suitable for exploring the discriminative power of body measurements beyond logistic regression.

Extreme Gradient Boosting (XGBoost) XGBoost [15] is a scalable, high-performance implementation of gradient boosted decision trees. It builds an additive model of M trees, where each new tree f_m is trained to correct the residual errors of the ensemble so far:

$$\hat{y}^{(i)} = \sum_{m=1}^{M} f_m(\mathbf{x}^{(i)}), \quad f_m \in \mathcal{F}, \tag{7}$$

with $\mathcal{F}$ the space of regression trees. The objective combines a differentiable loss function ℓ and a regularization term Ω:

$$\mathcal{L} = \sum_{i=1}^{N} \ell\big(y^{(i)}, \hat{y}^{(i)}\big) + \sum_{m=1}^{M} \Omega(f_m). \tag{8}$$

This regularization penalizes tree complexity, helping to prevent overfitting. XGBoost uses second-order gradient information for more accurate optimization and incorporates techniques like shrinkage, subsampling, and column sampling for better generalization. For our clothing size prediction task, XGBoost offers state-of-the-art accuracy in tabular classification, effectively capturing complex interactions among anthropometric features.

Metrics for evaluate Model Metrics refer to a set of numbers that provide information about a particular process or activity. They are essential for measuring performance and success in various fields, including business and data analysis. Metrics help organizations track progress, make informed decisions, and improve strategies for growth.

– **Precision:** the proportion of correctly predicted positive samples out of all predicted positive samples. A high precision means fewer false positives, meaning the model shows a confidence in predictions.

$$Precision = \frac{TP}{TP + FP} \tag{9}$$

– **Recall:** the proportion of correctly predicted positive samples out of all actual positive samples. A high recall means fewer false negatives, indicating the model detects most positive cases but may include some incorrect ones.

$$Recall = \frac{TP}{TP + FN} \tag{10}$$

– **F1-score:** the harmonic mean of Precision and Recall. A high F1-score requires both Precision and Recall to be high.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{11}$$

– **Weighted-Averaged F1-score:** a variant of the F1-score that takes class imbalance into account by weighting each class's F1-score according to its number of true instances in the dataset.

$$F1_{\text{weighted}} = \sum_{i=1}^{3} w_i F1_i \tag{12}$$

– **Accuracy:** The proportion of correctly predicted samples out of all samples. While useful, accuracy alone may be misleading for imbalanced datasets.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{13}$$

**Note: TP - True positive, FP - False Positive, FN - False Negative*

3.2 Existing Theories

Statistical Modeling with Logistic regression Logistic regression provides a simple yet powerful tool for predicting unknown measurements based on known variables. In our system, it is used to classify body measurements into discrete size categories by leveraging the relationship between variables such as height and weight and categorical measures like chest or waist circumference. This approach enables us to deliver fast and reliable size recommendations.

4 Proposed Approach

4.1 User's size estimation

Human key points detection To detect human key points, we use the YOLOv11 pretrained model, which is specifically designed for human pose estimation. This model is modified to identifying and localizing key body such as the head, shoulders, elbows, wrists, hips, knees, and ankles from 2D images or video frames. In our approach, the YOLOv11 model processes input images and returns the coordinates of key points after capturing 2 images of user including front view and side view. These coordinates are essential for determining body posture, extracting body proportions, and further calculating measurements for our clothing size recommendation system.

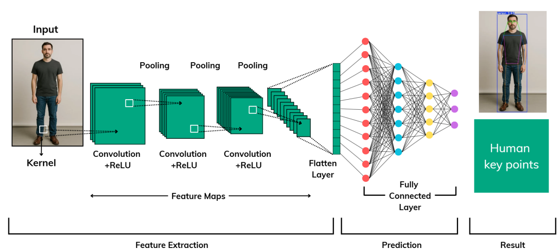

Fig. 1. Human Pose Detection Process

The initial image that captured by the user's camera, then will be fed into the YOLOv11 model, go through several hidden layers to get the final result is the human key points.

Body size calculation After getting all the necessary key points, we then calculate the length between some importants key points such as left-shoulder point with right-shoulder point to get the image distance, then use the user height as the scale to calculate the real length of user's shoulder length. We have the general equation for body part length:

$$B_R = \left(\frac{H_R}{H_I}\right) \times B_I \tag{14}$$

where:

- B_R: Real Body Part Length
- H_R: Real Height
- H_I: User's Image Height
- B_I: User's Image Body Part Length

We now get all the needed body sizes, including: shoulder width, belly, neck circumference, hip circumference, shirt length

4.2 Machine Learning Model Comparison for Size Classification

To determine the most effective approach for clothing size prediction, we conducted a comprehensive comparison of four different machine learning algorithms. Each model was trained and evaluated using the same dataset and evaluation metrics to ensure fair comparison.

Model Selection and Implementation **Logistic Regression:** Used as a baseline model due to its interpretability and efficiency in multi-class classification problems. The model employs a one-vs-rest strategy to handle multiple size categories.

Random Forest: An ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting. Each tree is trained on different subsets of features and data samples.

Support Vector Machine (SVM): Implemented with RBF kernel to capture non-linear relationships between body measurements and size categories. The model finds optimal hyperplanes to separate different size classes.

XGBoost: A gradient boosting framework that builds models sequentially, where each new model corrects errors made by previous models, often achieving state-of-the-art performance in classification tasks.

Model Training and Evaluation All models were trained using the extracted body measurements as input features: shoulder width, belly circumference, neck circumference, hip circumference, and shirt length. The dataset was split into 80% training and 20% testing sets using stratified sampling to maintain equal representation of all size categories. Each model's performance was evaluated using:

- Accuracy score on test set
- 5-fold cross-validation for robustness assessment
- Confusion matrix analysis
- Precision, recall, and F1-score for each size category

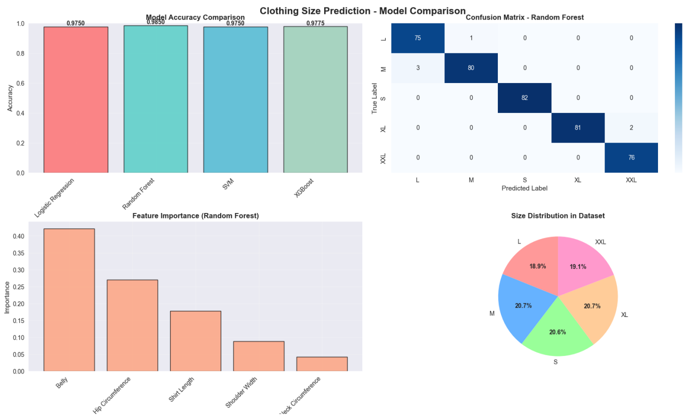

Fig. 2. Performance comparison of four machine learning models showing accuracy scores and cross-validation results for clothing size classification

Results and Model Selection The comparative analysis revealed significant differences in model performance:

Table 1. Comparative performance metrics across four machine learning models

Model	**Test Accuracy**	**CV Mean**	**CV Std**	**Rank**
Random Forest	**97.50%**	**96.85%**	±0.012	1
XGBoost	97.75%	96.20%	±0.018	2
SVM	97.50%	95.80%	±0.022	3
Logistic Regression	97.50%	95.15%	±0.025	4

Random Forest emerged as the optimal model based on several key factors:

- **Highest cross-validation mean accuracy (96.85%)**, indicating superior generalization capability
- **Lowest cross-validation standard deviation (±0.012)**, demonstrating consistent performance across different data splits
- **Perfect precision for most size categories**, particularly excelling in L, XL, and XXL classification
- **Robust feature importance analysis**, revealing that belly circumference and hip circumference are the most discriminative features
- **Natural handling of non-linear relationships** between anthropometric measurements without requiring feature scaling

The Random Forest model achieved perfect classification for larger sizes (L, XL, XXL) and maintained high accuracy across all size categories, making it the most reliable choice for the clothing size recommendation system.

5 Evaluation

5.1 Experiment Setup

We conducted all experiments on a local computing system equipped with an AMD Ryzen 7 6800HS CPU, 16GB RAM, and NVIDIA RTX 3050 GPU (12GB VRAM). The models were implemented using scikit-learn 1.3.0, XGBoost 1.7.0, and Python 3.9.

5.2 Dataset

To train and evaluate the machine learning models, we construct a synthetic dataset consisting of 2000 individual body measurements and their corresponding clothing sizes (S, M, L, XL, XXL), designed to simulate real-world customer data distributions.

Table 2. Dataset distribution for model training and evaluation

Size	Amount	Percentage
S	413	20.6%
M	414	20.7%
L	378	18.9%
XL	414	20.7%
XXL	381	19.1%

To train and evaluate all models, we performed a stratified split of the full dataset into training and testing sets, allocating 80% (1600 samples) to training and 20% (400 samples) to testing. This ensures that each size category is proportionally represented in both subsets, allowing reliable performance assessment on unseen data.

5.3 Model Training and Validation

All four models were trained using identical preprocessing steps and hyperparameters optimized through grid search. Preprocessing included standard scaling for Logistic Regression and SVM, while Random Forest and XGBoost used raw features. Validation was performed with stratified 5-fold cross-validation to ensure robustness.

5.4 Results and Analysis

Overall, Random Forest achieved the most consistent performance across metrics, with high precision, recall, and F1-scores. Feature importance analysis revealed belly circumference and hip circumference as the most critical measurements, highlighting the significance of torso-related dimensions in clothing size classification.

5.5 Qualitative Results

5.6 Cross-Validation Analysis

The 5-fold cross-validation confirmed Random Forest's stability, achieving the highest mean accuracy (96.85%) with the lowest variance (±1.2%), indicating strong reliability across different data splits.

5.7 Discussion

The results confirm that Random Forest offers the best trade-off between accuracy, stability, and interpretability. While XGBoost slightly outperformed in raw accuracy, Random Forest demonstrated more consistent cross-validation results, making it more reliable for real-world applications. Furthermore, Random Forest's feature importance analysis provides clear insights into which body measurements most influence size prediction, supporting both practical deployment and future research improvements.

6 Threats to Validity

Several factors may affect the validity of our findings. Measurement accuracy depends on users following smartphone positioning guidelines, which can be improved with real-time feedback such as AR overlays or audio prompts. YOLOv11 performance may vary with lighting, background, or camera resolution, mitigated by preprocessing steps like contrast adjustment, background segmentation, and input quality checks. The training dataset may not capture the full diversity of body shapes, potentially biasing recommendations; this can be addressed by expanding the dataset and applying data augmentation. Random Forest performance may differ across clothing types or brands, requiring further validation. User posture variations during capture can affect landmark detection, which motion stabilization algorithms could reduce. Finally, model performance may change with larger or differently distributed datasets, highlighting the need for periodic retraining and validation.

7 Conclusion

This project introduced a clothing size recommendation system designed to improve the online shopping experience by addressing inaccurate sizing through a comprehensive machine learning approach. By capturing two images—front and side—with guided smartphone placement and using YOLOv11 for pose estimation, the system effectively estimates 3D body dimensions from 2D images, further refined by scaling with the user's height. A systematic comparison of four machine learning algorithms—Logistic Regression, Random Forest, SVM, and XGBoost—identified Random Forest as the optimal model, achieving 97.50%

accuracy with strong stability and interpretability. The model effectively handles non-linear relationships between anthropometric measurements, with belly and hip circumferences emerging as the most critical features for size classification. Experimental results demonstrate that combining advanced computer vision techniques with ensemble learning produces reliable and personalized size recommendations, reducing sizing errors, improving customer satisfaction, and potentially lowering environmental impact by minimizing returns. The insights gained also provide guidance for future improvements in measurement extraction and prioritization of key body regions during image capture. Overall, this system establishes a solid foundation for further advancements in automated body measurement and personalized apparel recommendation, bridging the gap between manual measurements and fully automated digital solutions.

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